

Pioneer 3200 – Printed / Composite Airframe Structural Calculation

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Purpose and assumptions

This document checks whether a 1:1 copy of the Makeflyeasy Pioneer 3200 can be built from printed and composite material instead of EPO foam, and what it would weigh. Short answer: equal strength is achievable, but a fully printed airframe is 2.0–2.4 kg heavier. A molded composite shell matches the original weight, and a hybrid build lands about 0.85 kg over.

- **Geometry:** identical to the original, taken from the official STEP files and the RealFlight model: same airfoil, same angles.
- **Loads:** same maximum take-off weight and the same flight and hover cases as the original PNP.
- **Systems:** same motors, ESCs, servos and batteries; only the structure changes.
- **Safety factors:** 1.5 on limit load, plus a 1.25 knock-down on printed-material strength for scatter and weak layer bonding.
- **Fidelity:** first-pass engineering (simple beam theory, panel buckling), meant to set the order of magnitude. FEA and physical load tests are required before flight.

Pioneer baseline data

All calculations assume an MTOW of 24.9 kg on a 1.0 m² wing, a wing loading of 24.9 kg/m².

Parameter	Value	Source
Wingspan	3200 mm	KIT / PNP manual
Wing area	100 dm ²	Manual
Fuselage length / height	1680 / 410 mm	Manual
Root chord / airfoil thickness	~330 / ~45–52 mm	Measured on the model
KIT weight (airframe)	4.5 kg	KIT manual
PNP weight	9.2 kg	PNP manual

Parameter	Value	Source
MTOW	24.9 kg (at 500 m altitude)	RCDrone
Max payload	5 kg	RCDrone
Speeds: stall / cruise / max	14 / 21–22 / 35 m/s	RCDrone
Endurance	230 min with 1 kg, 176 min with 5 kg	RCDrone
Motors	X6525 pusher (563 g), four M6518 VTOL (423 g each)	PNP manual
Wing spar	2 carbon tubes 20×17×945 mm	KIT manual
Fuselage tubes	2 × 21×20×387; nose 12×10 and 10×8 ×280; tail 10×8×146 mm	KIT manual
VTOL boom station	±490 mm from the fuselage centreline	Measured on the model

Existing airframe weight breakdown

The EPO foam in the whole aircraft weighs only **~1.7 kg** of the 4.5 kg KIT. The other ~2.8 kg is carbon tubes, booms, plywood plates, plastic parts, landing gear and hardware, and a printed or composite build replaces only the foam share.

Part group	Foam volume (L)	Wetted / outer area (m²)	EPO weight at 35 g/L
Fuselage: nose, payload bay, mid, rear, hatch	17.3	1.04	~0.61 kg
Wings + ailerons	26.0	1.87	~0.91 kg
Tail (stabilizers, elevators, fin, rudder)	4.2	0.52	~0.15 kg
Total foam	47.5	3.43	~1.66 kg (1.4–2.1 kg at 30–45 g/L)
Rest of KIT (carbon, plywood, plastic, booms, gear)	–	–	~2.8 kg

Volumes were computed by voxelizing the STEP files (fuselage) and the RealFlight model (wing and tail), because the meshes are not closed. Fuselage foam walls come out 15–21 mm thick, and the wing is solid foam. The EPO density is my assumption, not a manufacturer figure.

Candidate materials

Every printable filament is 30–40 times denser than EPO, so a printed wall must be 20–40 times thinner than the foam wall it replaces. That is why buckling, not material strength, drives the design.

Material	Density (g/cm ³)	E (GPa)	Tensile XY (MPa)	Tensile Z (MPa)	HDT (°C)	Note
EPO (original)	0.03–0.045	~0.015	~0.4	–	~80	Absorbs impacts, easy to repair
LW-PLA (foaming)	0.5–0.8	1–1.5	15–20	~8	~55	Very light; softens in sun
PLA	1.24	3–3.5	50–60	20–30	~55	Not for field use
CF-PETG	1.29–1.32	4–6	45–55	~20	75–80	Easy to print; hardened nozzle
PA-CF (PA6 / PA12)	1.12–1.20	5–8	70–100	30–45	120–180	Best structural filament; dry and anneal
ASA	1.07	~2	~40	~20	~95	UV resistant, for outer skins
Carbon / epoxy (woven, vacuum)	1.5–1.6	45–60	400–600	–	80–120	Composite route
Glass / epoxy	1.8–1.9	18–25	250–350	–	80–120	RF-transparent, around antennas
Pultruded carbon strip / tube	1.55	110–135	1500–2000	–	~120	Spars and stringers

Typical values from manufacturer datasheets, approximate. The calculations use the low end and a further $\div 1.25$ for printed material. Check the datasheet of the actual filament you choose.

Load cases

The sizing case is a flight pull-up at an ultimate **6 g × 24.9 kg**. VTOL hover reverses the wing-root moment, but it is 3.3 times smaller.

Case	Assumption	Design value
Flight load factor	limit $n = +4.0$, safety factor 1.5	ultimate $n = 6.0$
Dive speed	$VD = 1.25 \times 35 = 43.75 \text{ m/s}$	$q = \frac{1}{2}\rho V^2 = 1172 \text{ Pa}$
VTOL hover	limit $n = 2.0$ (transition, gust, braking), $\times 1.5$	$n = 3.0 \rightarrow 366 \text{ N per boom}$
Tail load	$CL = 1.0$ on 0.196 m^2 at VD	229 N limit
Wing torsion	$C_m = -0.1$, chord 0.30 m	22.5 N·m ultimate at the root
Printed-material factor	scatter, layer bonding, creep	strength $\div 1.25$

Lift is taken as elliptical over the 1.6 m semi-span, with no relief from wing self-weight (conservative). The bending moment at station y_0 is:

$$M(y_0) = \int \text{from } y_0 \text{ to } b/2 \text{ of } w(y) \cdot (y - y_0) dy$$

$$w(y) = w_0 \cdot \sqrt{1 - (2y / b)^2}, \text{ scaled so that } \int \text{from } 0 \text{ to } b/2 \text{ of } w(y) dy = n \cdot W \cdot g / 2$$

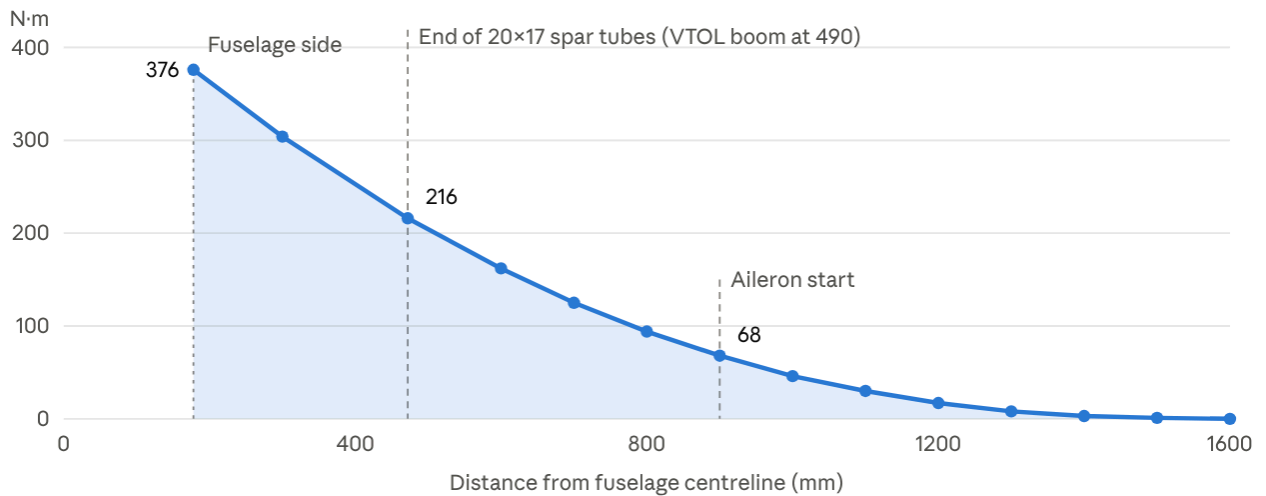
where $b = 3.2 \text{ m}$ span, $W = 24.9 \text{ kg}$, $g = 9.81 \text{ m/s}^2$ and $n = 6.0$.

Wing analysis

The two existing 20×17 carbon tubes carry the wing root at ~500 MPa ultimate and stay unchanged in every build route. Outboard of the tube end (472 mm) the printed wing needs its own spar, because a thin skin cannot carry 216 N·m.

The spar tube ends where most of the moment remains: the outer wing needs its own spar

At 472 mm the ultimate moment is still 216 N·m, 57% of the 376 N·m at the fuselage side



Calculated for this document · elliptical lift, $n = 6.0$, MTOW 24.9 kg

Item	Calculation	Result	Decision
Root spar	$M = 376 \text{ N}\cdot\text{m}$, $Z = 2 \times 375 \text{ mm}^3$	$\sigma \approx 500 \text{ MPa}$	Keep the original $2 \times 20 \times 17$ tubes
Outboard spar (from 472 mm)	$M = 216 \text{ N}\cdot\text{m}$, spar depth 40 mm, $\sigma = 600 \text{ MPa}$	9 mm^2 per cap	Pultruded carbon strip $10 \times 1 \text{ mm}$ top and bottom + printed web, tapering to the tip
Alternative: tube instead of strips	20×18 tube at 472 mm	$\sigma \approx 800 \text{ MPa}$	Marginal; strips work better because they sit far from the neutral axis
Skin – torsion shear	$T = 22.5 \text{ N}\cdot\text{m}$, 0.45 mm wall	$\tau \approx 3 \text{ MPa}$	Material is strong enough; buckling is the problem
Skin – shear buckling	$\tau_{cr} = 5.35 \cdot E \cdot (t / b)^2$	Ribs every 60 mm: 0.4–1.8 MPa ✗. 30 mm geodesic cells: CF-PETG 4.8, PA-CF 7.2 MPa ✓, LW-PLA 1.4 MPa ✗	Internal geodesic lattice $\leq 30 \text{ mm}$; for LW-PLA cells $\leq 20 \text{ mm}$ or a 0.6 mm wall
VTOL boom mount	hover $n = 3$: 366 N, 114 N·m at the root	Below the flight case	Double PA-CF rib around the boom, tied to both tubes

Hover loads act opposite to flight loads, so both spar caps work in compression as well as tension. Pultruded carbon in compression is roughly half as strong as in tension, which is why the calculation uses 600 MPa, not 1500.

Fuselage analysis

A 0.8 mm printed shell is strong enough in stress but buckles. The fuselage therefore needs a 1.0 mm PA-CF shell stiffened by an internal isogrid (or stringers every ~40 mm with frames every ~80 mm), and the two main carbon tubes upgraded from 21×20 to 22×20.

Design moment at the wing station: tail load $229\text{ N} \times 1.5$ on a 0.95 m arm, plus the inertia of 1.2 kg (tail, rear fuselage, servos) at 6 g on a 0.55 m arm, giving **$M \approx 383\text{ N}\cdot\text{m}$ ultimate**. The section is taken as a thin-walled ellipse 150 mm wide (2a) and 200 mm high (2b).

$$I \approx (\pi/4) \cdot t \cdot b^2 \cdot (3a + b)$$

$$\sigma = M \cdot b / I$$

Shell buckling: $\sigma_{cr} \approx 0.3 \cdot E \cdot t / R$ (0.3 instead of the classical 0.6 to cover imperfections)

Flat panel between stringers: $\sigma_{cr} \approx 3.6 \cdot E \cdot (t / bs)^2$

where t is the wall thickness, R the local radius (~100 mm) and bs the stringer spacing.

Shell configuration	Bending σ (MPa)	Allowable $\div 1.25$ (MPa)	Buckling (MPa)	Result
CF-PETG, 0.8 mm, no stringers	18.8	36	9.6	✗ buckles
CF-PETG, 1.2 mm, no stringers	12.5	36	14.4	Marginal
PA-CF, 0.8 mm, no stringers	18.8	56	14.4	✗ buckles
PA-CF, 1.0 mm + stringers every 40 mm	~15	56	13.5 (panel) + tubes	✓ with 2 × 22×20
Tubes alone: 2 × 21×20 (original)	1188	~600	–	✗ the original relies on the foam
Tubes alone: 2 × 22×20	578	~600	–	✓ marginal; with the shell ✓

- **Openings:** the payload-bay top hatch interrupts the upper shell. It needs a 3 mm surrounding frame with a 15 mm overlap.
- **Load points:** landing gear, wing attachment and the pusher motor sit on solid PA-CF frames (100% infill, 4 mm) with metal inserts.
- **VTOL landing:** a 1.5 m/s touchdown on stiff legs gives 6–8 g at the attachment. EPO forgives this and a printed shell cracks, so use flexible carbon legs or TPU shock absorbers.

Tail, VTOL booms and motor mounts

The tail is where every gram costs most. A CF-PETG tail is 0.31 kg heavier than the original, and balancing it takes about 0.6 kg of extra nose ballast 0.5 m ahead of the CG.

Item	Requirement	Proposed solution
Stabilizers and elevators	229 N limit on 0.196 m ² ; flutter margin at 44 m/s	LW-PLA or composite over a foam core; the 10×8 joiner tube stays
Vertical fin	Hosts the UHF antenna; must be RF-transparent	No carbon and no CF filament near the antenna: ASA or PETG, or glass fibre
VTOL booms	423 g motor + 22" prop; up to 15 kg thrust per motor	Keep the original carbon tubes; print only the fairing
VTOL motor mounts	Vibration at ~3,000–5,000 rpm, heat from the motor	Annealed PA-CF or aluminium; never PETG or PLA
Pusher motor mount	563 g, up to ~4 kW, 21" prop	Carbon or G10 plate on a PA-CF frame

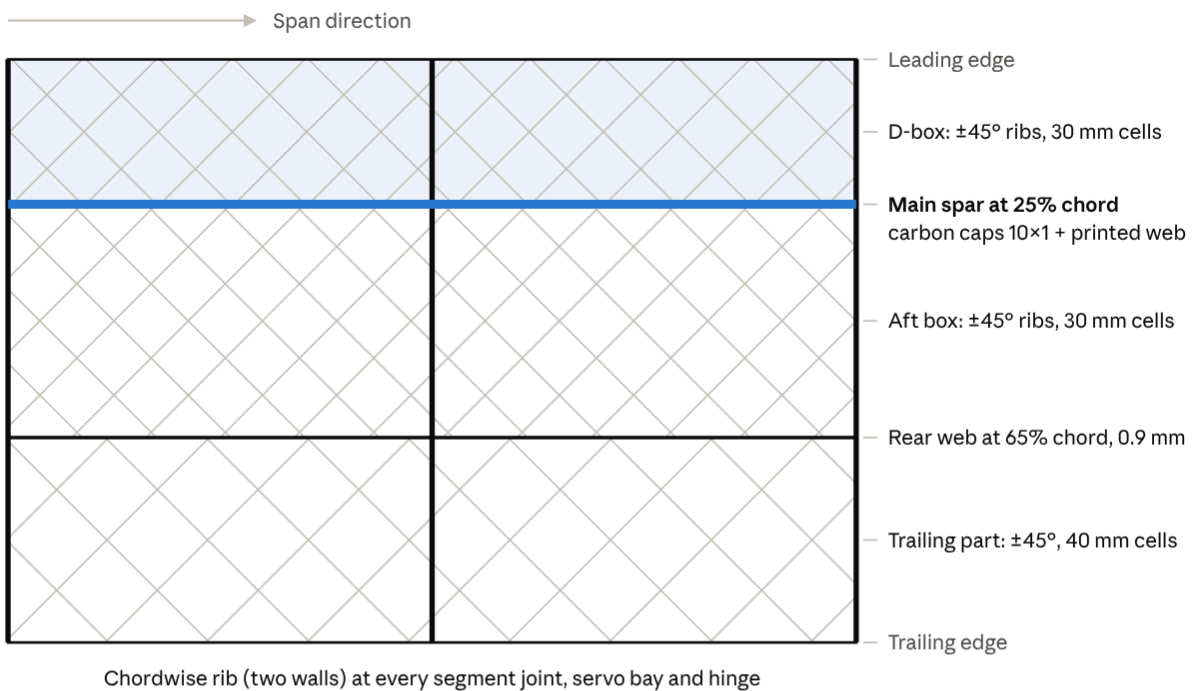
Motor vibration is the main fatigue driver in printed parts. EPO damps it, while a stiff printed structure passes it on. Use rubber isolators between boom and wing and around the flight controller.

Printed section design: directional internal ribs

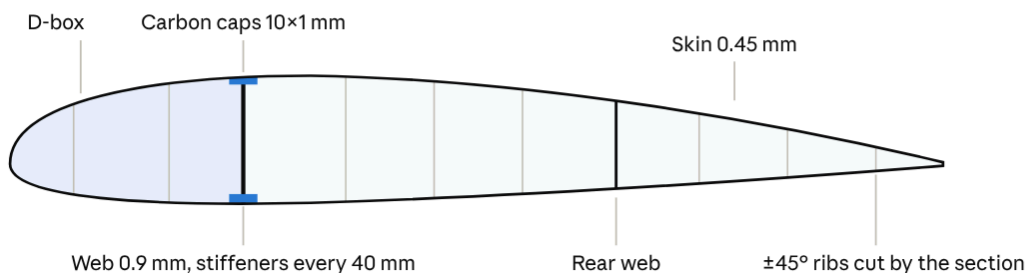
Each internal rib follows the direction of the stress it carries. Spanwise carbon caps take bending, a $\pm 45^\circ$ lattice takes torsion shear, chordwise ribs hold the shape and take local crushing, and the fuselage uses a $0^\circ/\pm 60^\circ$ isogrid. Sized this way, the wing skin needs 30 mm cells and the fuselage 35 mm triangles.

Wing: carbon caps carry bending, $\pm 45^\circ$ ribs carry torsion, chordwise ribs hold the shape

Plan view: two 240 mm segments, leading edge at the top



Section through the segment (chord 330 mm, thickness exaggerated $\times 1.6$)



Inboard of 472 mm the caps end in sockets over the two $\varnothing 20 \times 17$ spar tubes, with a 100 mm overlap.

Proposed rib layout · outboard wing segment, plan and section

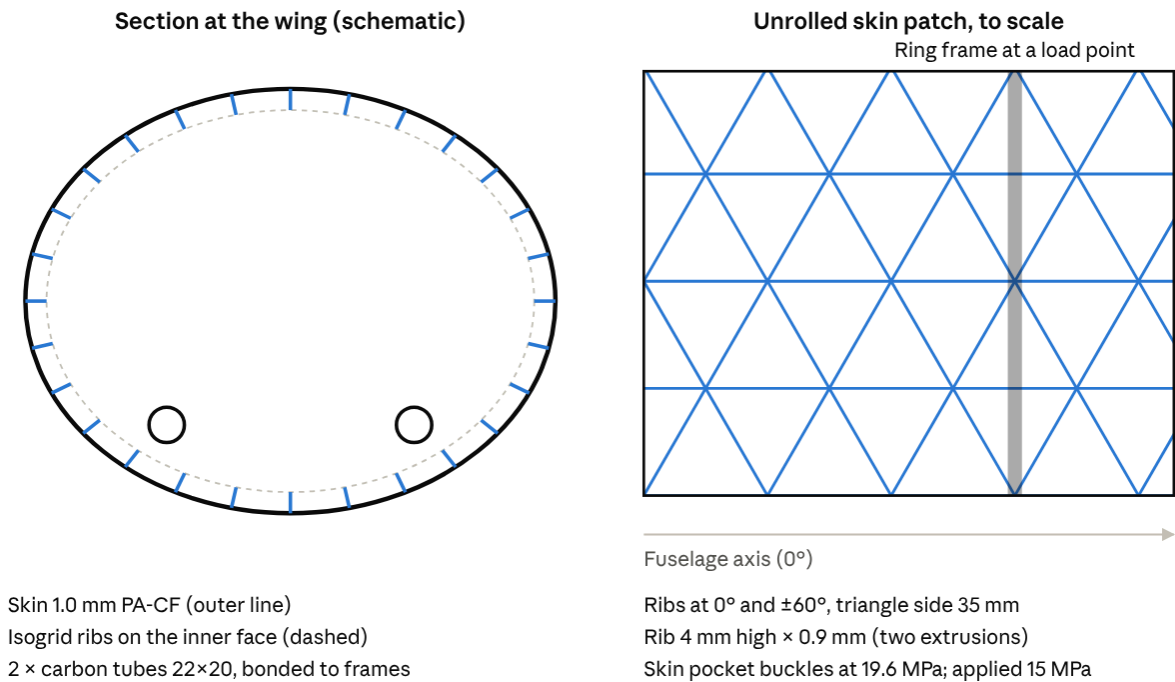
The lattice divides the 0.45 mm skin into small panels that cannot buckle. Because torsion produces principal stresses at $\pm 45^\circ$, the diagonal ribs also carry that load directly in tension and compression.

Why each direction:

- **0° (spanwise or along the fuselage):** bending puts the top and bottom of the section into tension and compression along the axis. The most efficient material is a continuous carbon strip as far from the neutral axis as possible, which is why the caps sit on the skin.
- **$\pm 45^\circ$:** torsion and transverse shear produce principal stresses at 45° to the axis. Ribs on those lines carry them as a truss and break the skin into panels that stay stable.

- **90° (chordwise ribs, ring frames):** keep the airfoil or section shape, spread point loads and close each segment at its joint.
- **Radial fans:** at concentrated loads (boom bolts, servo screws, gear) ribs fan out from the insert into the lattice, with 100% infill within 15 mm of the insert.

Fuselage: a 0°/±60° isogrid stiffens a 1.0 mm PA-CF skin against buckling



Proposed rib layout · fuselage section and isogrid patch

Region	Dominant load	Rib direction	Spacing	Thickness
Wing D-box (leading edge to spar)	Torsion shear	±45° geodesic	30 mm cells	0.45 mm
Wing spar	Bending	0° carbon caps + printed web with vertical stiffeners	stiffeners every 40 mm	web 0.9 mm
Wing aft box	Shear, shape	±45°	30 mm	0.45 mm
Trailing part	Light shear	±45°	40 mm	0.45 mm
Segment joints, servo bays, hinges	Crushing, local loads	90° chordwise rib	every 240 mm + at each fitting	0.9 mm

Region	Dominant load	Rib direction	Spacing	Thickness
VTOL boom mount	366 N point load	radial fan + solid ring	–	solid
Fuselage shell	Bending + torsion + pressure	isogrid $0^\circ/\pm 60^\circ$	35 mm triangles	skin 1.0, ribs 4×0.9 mm
Fuselage load points	Concentrated	ring frames + longerons to the tubes	at each load point	3 mm
Fin and stabilizers	Bending, flutter	spanwise spar + $\pm 45^\circ$	30 mm	0.45 mm

Sizing checks ($E = 4$ GPa for CF-PETG, 6 GPa for PA-CF):

Check	Formula	Value	Limit	Result
Spar web shear at 472 mm	$\tau = V / (h \cdot t) = 461 \text{ N} / (40 \text{ mm} \times 0.9 \text{ mm})$	12.8 MPa	20 MPa (CF-PETG 25 ÷ 1.25)	✓
Web shear buckling, no stiffeners	$\tau_{cr} = 5.35 \cdot E \cdot (t / h)^2$	10.8 MPa	must exceed 12.8	✗
Web with stiffeners every 40 mm	$\tau_{cr} = 9.34 \cdot E \cdot (t / h)^2$	18.9 MPa	must exceed 12.8	✓
Cap-to-tube bond length	$F = M / h = 5.4 \text{ kN}$; $L = F / (\tau_{\text{bond}} \cdot p)$ with 10 MPa and 12 mm perimeter	45 mm	use 100 mm	✓ ×2 margin
Wing skin cell, CF-PETG 0.45 mm	$\tau_{cr} = 5.35 \cdot E \cdot (t / b)^2$, $b = 30 \text{ mm}$	4.8 MPa	must exceed 3.2	✓
Fuselage skin pocket, PA-CF 1.0 mm	$\sigma_{cr} \approx 4 \cdot E \cdot (t / a)^2$, $a = 35 \text{ mm}$	19.6 MPa	must exceed 15	✓
Isogrid added mass	$(2\sqrt{3} / a) \cdot h_r \cdot t_r = 0.099 \times 4 \times 0.9$	+0.36 mm equivalent skin	1.56 kg/m ² total	Matches the weight budget

Slicer and modelling notes:

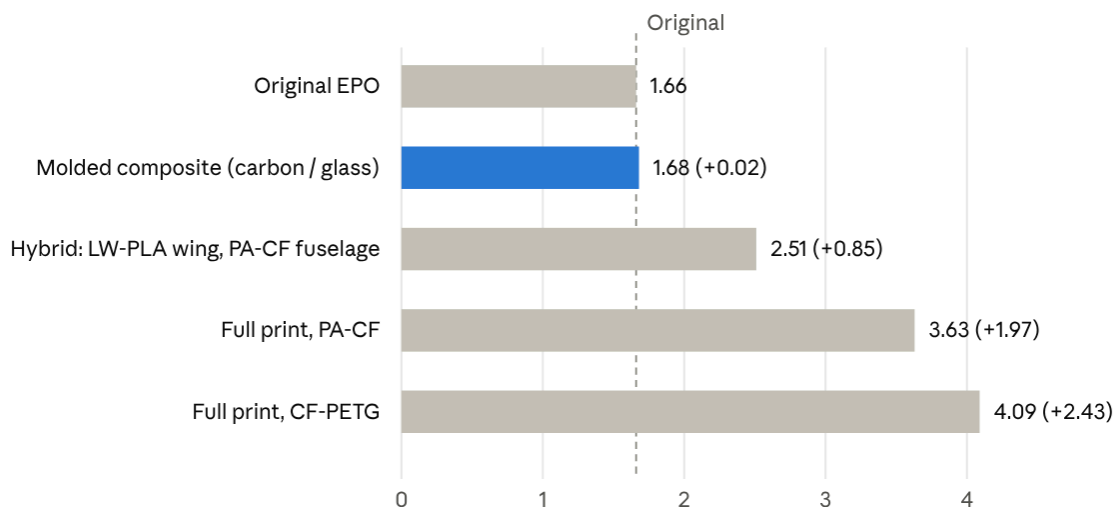
- Model every rib as a wall exactly one or two extrusion widths thick (0.45 or 0.9 mm), so the slicer prints it as one continuous path fused to the skin, with no infill.
- Print wing segments standing on the root rib (span along Z). Bending goes to the carbon caps, so the weak Z direction sees only light skin loads, while the ±45° ribs carry shear within each layer.
- Print fuselage halves lying down, split at the waterline, so longitudinal bending runs in the XY plane.
- Lightening holes in chordwise ribs up to 50% of rib height with 3 mm flanges; route cables through the ribs near the neutral axis.

Comparative weight budget

A molded composite shell matches the original weight, the hybrid route adds ~0.85 kg, and a full print adds 2.0–2.4 kg. Areas are 1.04 m² of fuselage shell and 2.39 m² of wetted wing, aileron and tail surface. Each route also gets 0.1 kg of outboard spar caps.

Only a molded composite shell keeps the original weight; a full print adds 2–2.4 kg

Weight of the skins and their internal structure, kg. The other ~2.8 kg of the KIT is the same in every route.



Calculated for this document · areas from the model, areal densities from the table below, EPO at 35 g/L

Route	Wing and tail (kg/m ² wetted)	Fuselage (kg/m ²)	Build-up
Full print, CF-PETG	0.88	1.82	0.45 mm wall + geodesic lattice; fuselage 1.0 mm + stringers
Full print, PA-CF	0.78	1.61	Same, density 1.15
Hybrid	0.45 (LW-PLA)	1.29 (PA-CF 0.8 mm + stringers)	Foamed wing, structural fuselage
Molded composite	0.40	0.60	2 plies of carbon / glass at 80–160 g/m ² + local core, in printed moulds

All densities include a 40–50% allowance for internal structure (ribs, stringers, joints). A printed frame can replace some of the original plywood and save another 0.1–0.2 kg, which is not counted here.

Performance impact

The official figures (230 min with 1 kg, 176 min with 5 kg) give a sensitivity of **~13.5 minutes of endurance per kg**, about 6% per kg. MTOW is fixed at 24.9 kg, so every gram added to the structure comes straight off the payload.

Build route	Added weight	Endurance with 1 kg payload	Max payload
Original EPO	–	230 min	5.0 kg
Molded composite	+0.02 kg	~230 min	~5.0 kg
Hybrid	+0.85 kg	~219 min	~4.15 kg
Full print PA-CF	+1.97 kg	~203 min	~3.0 kg
Full print CF-PETG	+2.43 kg	~197 min	~2.6 kg

- **Centre of gravity:** extra weight in the tail needs nose ballast (0.6 kg for full print), which is not included above. Moving the batteries forward can replace the ballast.
- **Hover:** the VTOL motors (up to 15 kg thrust each) can absorb even +2.4 kg as long as MTOW holds. Hover power scales with roughly $(W'/W)^{1.5}$, so 2 kg extra costs about 7% more hover power.
- **Stall speed:** scales with $\sqrt{W'/W}$. At a fixed take-off weight it does not change, because the structure takes the payload's share.

Manufacturing

The large fuselage parts are 374–388 mm long, so they fit whole only on a 400 mm bed. On a 256 mm printer each part splits in two, and each wing half splits into segments of ~240 mm.

Part	Size (mm)	Pieces on a 256 mm printer	Pieces on a 350–400 mm printer
Nose	374 × 272 × 240	2	1
Payload bay	388 × 336 × 243	2–3	1 (on a 400 bed)
Rear fuselage	380 × 151 × 207	2	1
Vertical fin	281 × 51 × 263	2	1

Part	Size (mm)	Pieces on a 256 mm printer	Pieces on a 350–400 mm printer
Wing half	1408 × 330 × 52	6 segments	4 segments
Stabilizer + elevator	380 × 220	2	1

Recommended print settings:

- **Wing and tail skins:** one 0.45 mm wall with a 0.4 mm nozzle, no infill. The geodesic lattice is modelled in CAD and printed as a single wall, in the 3DLabPrint / Eclipson style.
- **PA-CF fuselage:** 0.6 mm hardened nozzle, 0.2 mm layers, 2 perimeters. Use an enclosed printer with an 80–100°C bed, dry the filament before printing (80°C, 8 h) and anneal afterwards.
- **Print orientation:** orient layers so that bending stress runs in the XY plane, not across layers. For the fuselage that means printing the halves lying down, split at the waterline, so the long axis lies in XY.
- **Joints:** 15 mm lap joints with thickened epoxy, or sleeves sliding over the carbon tube. The spar runs through all segments and aligns them.

Composite route: print the moulds, not the aircraft. Outer moulds in ASA or PETG are generated from the existing STEP files (outward offset), sanded and sealed, then laid up with carbon and glass in a vacuum bag. A second option is hot-wire-cut XPS cores sheathed in glass, which suits the wings and tail especially well.

Material and time for a full print: about 4 kg of filament and over 100 print hours at 30–40 g/h. At USD 40–80 per kg of PA-CF, filament alone costs about USD 160–320.

Risks, validation and recommendation

Recommendation: build a composite shell in printed moulds and print only the internal frames and mounts in PA-CF. It is the only route that matches both the strength and the weight of the original. If ease of manufacture matters most, the hybrid route (+0.85 kg) is a reasonable compromise.

Risk	Effect	Mitigation
Layer separation (Z direction)	Sudden skin crack	Correct print orientation, 1.25 stress knock-down, annealed PA-CF

Risk	Effect	Mitigation
Heat and sun	LW-PLA and CF-PETG soften; airfoil distorts	White paint; ASA or PA-CF for outer skins
Impact and landing damage	EPO absorbs, printed parts crack	Flexible legs and TPU dampers, a glass layer on the fuselage belly
Vibration and fatigue	Cracks around motor mounts	Isolators, fillet radii ≥ 3 mm, vibration test
RF blocking	CF filament conducts and detunes the antenna	Carbon-free fin and antenna bays
Centre of gravity	Heavy tail needs ballast	LW-PLA or composite tail, batteries forward

Validation tests before flight:

- ☐ Coupon tests: XY and Z tensile strength of the actual filament and settings
- ☐ Sandbag wing test to 4 g (limit): no permanent deformation. To 6 g (ultimate) on a test wing only
- ☐ Wing torsion and stiffness test (aileron flutter margin at 44 m/s)
- ☐ Tail load test: 340 N on the stabilizers, fuselage clamped at the wing
- ☐ Drop test at full weight from 0.3 m
- ☐ Tethered hover with vibration logging (ArduPilot IMU log)

All calculations here are first-pass (beam theory, panel buckling, estimated densities). FEA on the final geometry and coupon tests are required before a prototype.

Sources and assumptions

- [Makeflyeasy MFE_ArduPlane – Pioneer folder](#): KIT and PNP manuals, STEP files, RealFlight model
- [RCDrone – Makeflyeasy Pioneer 3200mm product page](#): MTOW, speeds, endurance, payload
- Assembled model `pioneer/pioneer_full.glb` in the ohameiri/maya repository: dimensions, areas and volumes

My assumptions, not manufacturer data:

- EPO density 35 g/L (range 30–45)
- Load factors $n = 4$ in flight and 2 in hover, $\times 1.5$
- Allowable 600 MPa for pultruded carbon in compression

- Typical filament properties from manufacturer datasheets
- $C_m = -0.1$ and tail $CL = 1.0$
- A 40-50% allowance for internal structure

The Python script that runs these calculations is in the working environment and can be re-run with other assumptions.